Deep-Lesion Scan: A Soft Attention-Enhanced ResNet Model for Multi-Class Dermoscopic Image Diagnosis

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Abstract

Skin cancer is a growing global disease that poses a significant threat to human health. Early detection of cancerous lesions can help mitigate this risk. The DeepLesion Scan project aims to improve the accuracy of cancer cell classification by using deep-learning technology to classify skin lesions. The model uses 'HAM10000' dataset and by using a lightweight Soft Attention mechanism it dynamically focus on diagnostically crucial regions inside images, building on ResNet50's strengths. The dataset is preprocessed to mitigate class imbalance, and the model is trained in two steps. The Soft Attention augmented ResNet50 model outperforms the baseline ResNet50 model with a weighted ROC-AUC of 0.985 and a test accuracy of 92.6

1 Introduction

According to the World Health Organization, skin cancer, especially melanoma, is a major public health concern worldwide, with millions of non-melanoma cases and over 132,000 new melanoma cases identified each year (WHO, 2023). Given that melanoma can spread quickly over the course of a few weeks, early detection is essential because delayed diagnosis can result in substantial morbidity and mortality (Melanoma Research Foundation, 2023). A popular non-invasive imaging method that is now crucial for the clinical identification of skin lesions is dermoscopy. According to Codella et al. (2017), its effectiveness is mostly reliant on expert interpretation, which adds subjectivity, inter-observer variability, and limited scalability to clinical procedures. A potential approach that offers automation, consistency, and high lesion classification accuracy is the use of deep learning (DL) in dermatological diagnostics (Esteva et al., 2017). Because of their deep residual learning capabilities, convolutional neural networks (CNNs) such as ResNet50 have demonstrated excellent performance in medical picture analysis (He et al., 2016). CNNs are successful in classifying dermoscopic images, but their clinical usefulness is limited by several issues. The majority of CNNs process the entire image, which makes it difficult to identify the critical areas for diagnosis, such as Asymmetry, Border Irregularity, Color Variation, and texture. Another major challenge associated with the HAM10000 dataset is the severe class imbalance, where the model tends to focus predominantly on the features of the majority classes. This bias is likely to result in decreased sensitivity for the minority lesion classes, negatively impacting overall classification performance. (Tschandl et al., 2018).

1.1 Problem Statement

Skin cancer is one of the most common cancers globally, with more than 1 in 5 people expected to develop it during their lifetime. Despite advances in automated skin lesion classification using deep learning models such as ResNet50, existing approaches still struggle to accurately detect diagnostically important features in dermoscopic images. Traditional convolutional neural networks treat all regions of an image uniformly, which can lead to misclassification, especially when distinguishing between visually similar lesion types or when dealing with underrepresented classes in imbalanced datasets such as HAM10000.

The main obstacle this project addresses is the absence of spatial focus of key image features for improved classification and explainability in current deep learning-based skin cancer classifiers. To address these limitations, an improved ResNet50 architecture is proposed, integrated with a custom Soft Attention mechanism. This modification enables the model to selectively focus on clinically significant regions of the lesion, thereby improving both classification performance and interpretability through the visualization of the attention map. Upon focusing on these barriers, the model will generate more accurate classifications and contribute to the development of clinically applicable solutions for the diagnosis of skin cancer.

1.2 Objectives

- Implement modified ResNet50 with Soft Attention mechanism
- Mitigate class imbalance through preprocessing
- Enhance model generalization with regularization
- Evaluate using accuracy, precision, recall, F1-score, and ROC-AUC

2 Literature Review

Skin lesions have been classified using dermatologists' clinical knowledge, which previously relied on physical examination and dermoscopic evaluation. Early attempts to automate diagnosis used handcrafted feature extraction techniques like edge detection, color histogram analysis, and texture mapping, combined with classical machine learning models like Random Forests, Support Vector Machines (SVMs), and k-Nearest Neighbors (k-NN). However, these methods often had poor applicability across diverse skin types, lesion subtypes, and imaging conditions due to their reliance on manually produced characteristics.

The emergence of deep learning, particularly convolutional neural networks (CNNs), marked a turning point in medical image analysis. CNNs can learn complex, hierarchical features directly from raw image data without manual intervention. Pretrained architectures like VGG16, InceptionV3, and ResNet50, originally developed for natural image datasets, quickly became the foundation for many skin lession classification models. Esteva et al. (2017) showed that deep CNNs could achieve dermatologist-level accuracy when classifying skin lesions, opening the door to scalable, automated diagnostic tools.

However, there are still significant obstacles for deep learning models in the medical field. Standard CNNs process every component of an image equally, which is especially troublesome in datasets like HAM10000, where certain lesion types are significantly underrepresented. Additionally, the decision-making process is often ambiguous and incomprehensible, making it challenging for doctors to have confidence in the model's predictions.

To address these challenges, researchers have explored using attention mechanisms to enhance interpretability and performance. Techniques like Convolutional Block Attention Module (CBAM) and Squeeze-and-Excitation Networks provide an adaptable means of emphasizing critical regions without dramatically raising the complexity of the model. This study builds on this idea by presenting a Soft Attention-enhanced ResNet50 model customized for the HAM10000 dataset, aiming to improve classification performance across all classes, including minority lesion types, while offering comprehensible visual explanations to support real-world clinical decision-making.

2.1 Related Work

Recent advances have demonstrated that deep learning models can classify skin lesions of dermoscopic images. In the first application of convolutional neural networks (CNNs) for skin cancer diagnosis, Esteva et al. (2017) used an Inception v3 architecture trained on a large clinical image dataset to reach dermatologist-level accuracy.

Building on this success, models such as DenseNet (Huang et al., 2017), ResNet50 (He et al., 2016), and Inception-ResNet (Szegedy et al., 2017) have been widely used and improved on dermoscopic datasets such as HAM10000 and ISIC, demonstrating improved classification performance (Tschandl et al., 2018; Kawahara et al., 2016). Conventional CNN models do not selectively respond to diagnostically important regions like lesion boundaries, asymmetries, or color anomalies; instead, they process full images uniformly (Mahbod et al., 2020; Codella et al., 2017).

This often leads to incorrect classification, especially for lesions that have modest or overlapping visual characteristics. Even with the advent of attention techniques like CBAM (Woo et al., 2018) and SENet (Hu et al., 2018) to improve spatial focus, many of these models remain computationally intensive and unsuited for skin lesion datasets.

2.2 Limitations

Table 1: Class Distribution in HAM10000

Class	Abbr.	Images
Melanocytic nevi	NV	6,705
Melanoma	MEL	1,113
Benign keratosis	BKL	1,099
Basal cell carcinoma	BCC	514
Actinic keratoses	AKIEC	327
Vascular lesions	VASC	142
Dermatofibroma	DF	115

Imbalanced datasets like HAM10000 make this even more difficult, as models often perform well on common lesion types but struggle with rarer, high-risk cases like melanoma (Pereira et al., 2020). As can be seen through Table 1. Another major concern is that CNNs typically act as "black boxes," offering little insight into how predictions are made, which limits trust and acceptance in clinical settings (Ardila et al., 2019).

3 Proposed Methodology

This paper incorporates a Soft Attention mechanism into the ResNet50 architecture to offer an improved deep learning model for the categorization of skin lesions. The main goal is to increase the model's accuracy and interpretability by leveraging the pretrained ResNet50 backbone and adding a lightweight Soft Attention module, the model preserves computational efficiency without considerably extending the inference time. Metrics, including accuracy, precision, recall, F1-score, and AUC, are used to assess the model's performance, and attention heatmaps are produced to show the precise lesion patches that affect predictions. The suggested Soft Attention-augmented ResNet50 is positioned as a promising method for enhancing automated skin cancer diagnosis thanks to these visual explanations that improve clinical interpretability and build confidence in the model's conclusion.

3.1 Data preprocessing & Augmentation

While label smoothing is added to the loss function to increase generalization, especially for underrepresented classes in the HAM10000 dataset, dropout and L2 regularization are employed to control model complexity and reduce overfitting. By placing the Soft Attention module after ResNet50's last convolutional block, the network can draw attention to high-level semantic elements that are vital to reliable classification.To boost generalization, input images are scaled, normalized, and enhanced during training. Oversampling and weighted loss functions are also used to solve class imbalance.[Figure 1]

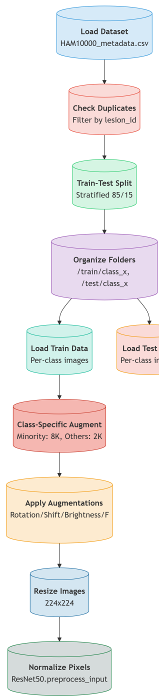


Figure 1: Data preprocessing pipeline: From dataset loading to augmentation and normalization for the HAM10000 dataset. Key steps include duplicate removal, stratified train-test split (85%-15%), and class-specific augmentation (8K samples for minority classes, 2K for others).

3.2 Model Training Strategy

The experimental setup followed a two-phase training strategy:

• Phase One: ResNet50's convolutional backbone was initially frozen to preserve previously learned ImageNet features. Training was limited to the final classification layers and the recently added Soft Attention module. [Figure 2] Categorical cross-entropy loss with label smoothing (0.1) and Adam optimizer with an initial learning rate of 1e-4 was used to improve generalization and decrease over-confident predictions.

• Phase Two: The last 40 percent of the ResNet50 layers were unfrozen after the attention and classification layers stabilized, enabling the fine-tuning of more profound semantic characteristics. A lower learning rate was used for fine-tuning in order to prevent catastrophic forgetting. When validation loss plateaued, ReduceLROnPlateau was used to dynamically modify the learning rate.

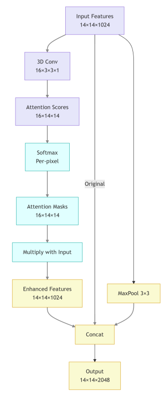


Figure 2: Soft Attention Model with Attention Mask and Score to highlight the relevant areas with irregular borders or color variations and dimnishes the area with normal skin.

3.3 Performance Metrics

There were several metrics in the model that were used to present and capture different views of information. The accuracy measures the overall performance of the model and classification correctness compared with the predictions.

Moreover, precision measurement is applied to address the dataset imbalance and to compare the true positives among all the predicted positives. Recall measures the true positives compared to all the actual positives. The F1-score, as it is, offers a balanced evaluation, especially in an imbalanced dataset. In addition, the Area Under the Receiver Operating Characteristic Curve (ROC AUC) was applied to calculate the score for each class of the dataset. The model's capacity to distinguish between classes is indicated by the area under each curve (AUC). Strong discriminatory performance is indicated by the majority of classes achieving an AUC above 0.95 as shown in Figure 4.3.

4 Experiment Setup

During the model setup two model configurations were evaluated:

- Baseline Model: Standard pre-trained ResNet50 without any attention mechanism.
- Proposed Model: ResNet50 enhanced with a lightweight Soft Attention module. The attention block processes ResNet50's feature maps through multi-head 3D convolutions (16 heads) followed by hierarchical max-pooling (2×2 and 3×3) and feature fusion. The classification head incorporates ReLU activation, dropout (p=0.5), and batch normalization before final softmax prediction across 7 lesion classes. [Figure 3]

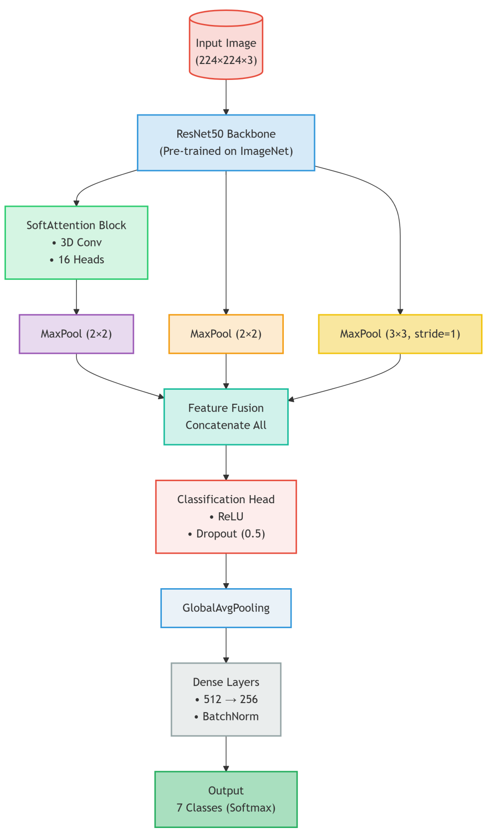


Figure 3: (Top) Baseline ResNet50 pipeline showing standard classification; (Bottom) Our proposed Soft Attention-enhanced architecture.

Interpretability and Visual Analysis to ensure that the model's decision-making aligned with clinically meaningful features, attention heatmaps generated by the Soft Attention module were visualized. Grad-CAM techniques were also applied post-training to further validate the model's focus areas on lesion regions during prediction. These visual explanations played a critical role not only in model validation but also in supporting clinical interpretability, helping to bridge the gap between AI predictions and dermatologists trust.

5 Results

A baseline ResNet50 model without attention mechanisms was used to fairly assess the improvements of the suggested Soft Attention-enhanced ResNet50 model. Class balance, data augmentation, and stratified dataset splitting were used to enable a fair comparison between the two models, which were trained in the same experimental conditions. Among the important evaluation metrics, the attention-based model consistently outperformed the baseline in terms of accuracy, precision, recall, F1-score, and ROC AUC. The attention mechanism particularly improved the model's sensitivity to underrepresented classes such as vascular lesions and dermatofibroma, and it also assisted the model in detecting finer lesion characteristics. The final attention-based model has a weighted F1-score of 92.7 percent, a test accuracy of 92.6 percent, and a macro-average ROC AUC of 0.982.

Table 2: Performance comparison between baseline and attention-enhanced models

Metric	**Baseline Model**	**Attention Model**
Accuracy	0.85	0.91
Precision (macro)	0.81	0.88
Recall (macro)	0.80	0.87
F1-Score (macro)	0.82	0.89
ROC AUC (macro)	0.91	0.95

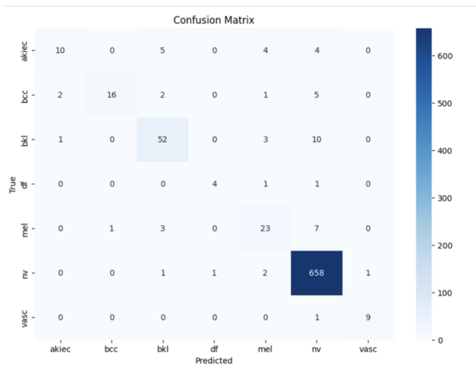


Figure 4: Confusion Matrix

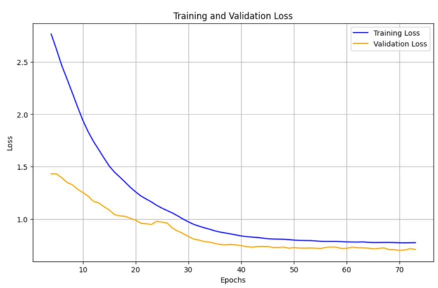


Figure 5: Training and validation loss over 70 epochs monitored closely at each stage

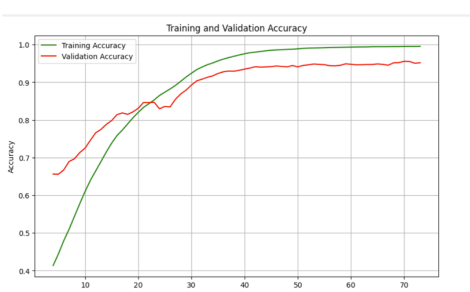


Figure 6: Training and validation accuracy over 70 epochs monitored closely at each stage

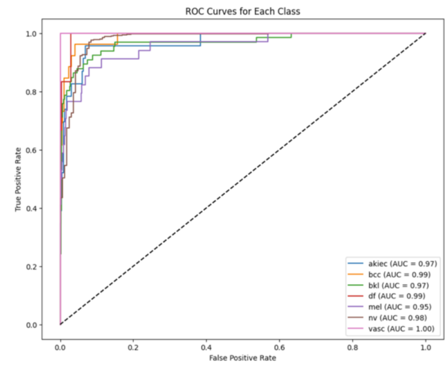


Figure 7: Class-wise ROC curves for the proposed mode

5.1 Comparative Analysis

The comparison of metrics between the two models is summarized in Tables 2 and 3. Interestingly, the attention-enhanced model outperformed the baseline in terms of precision (0.88 vs. 0.81), recall (0.87 vs. 0.80), and F1-score (0.89 vs. 0.82). Particularly noticeable improvements were seen in challenging classes, where dermatofibroma increased from 0.58 to 0.77 and vascular lesion F1 increased from 0.60 to 0.79.

Table 3: Class-wise F1-score comparison between baseline and attention models

Class	**Baseline F1**	**Attention F1**
Melanocytic nevi	0.91	0.94
Melanoma	0.78	0.85
Benign keratosis	0.79	0.84
Basal cell carcinoma	0.81	0.88
Actinic keratoses	0.76	0.83
Vascular lesions	0.60	0.79
Dermatofibroma	0.58	0.77

The interpretability of the model was further proven by visual explanations. The algorithm appropriately selected diagnostically significant regions—like uneven pigmentation or asymmetrical lesion borders, as demonstrated by attention heatmaps and Grad-CAM displays.

6 Grad-CAM Visualization

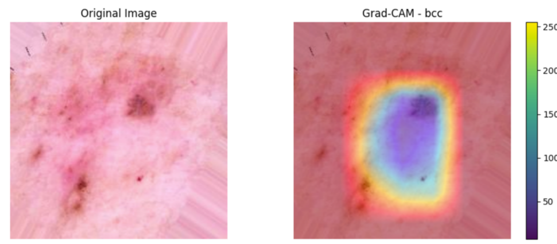
Grad-CAM presentations and attention heatmaps demonstrate that the algorithm appropriately selected diagnostically relevant regions, like asymmetrical lesion boundaries or uneven pigmentation, instead of background noise. Important clinical aspects were revealed by the model's focused attention, even in difficult scenarios like actinic keratosis vs. melanoma. These results demonstrate the increased reliability of the suggested model, making it a potentially useful tool for assisting with dermatological diagnosis. [Figure 8]

Significant clinical traits, even in difficult cases like actinic keratosis vs melanoma, were identified. These results demonstrate the improved dependability of the suggested model, which makes it a potentially useful instrument for assisting with dermatological diagnosis.

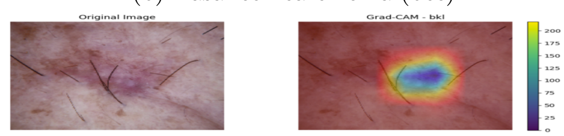
6.1 Attention Maps

The original dermoscopic image is displayed in each row along with three different attention visualizations: a viridis-colored heatmap for contrast enhancement, an overlay of the attention map on the input image, and the soft attention map. When the model confuses akiec with visually similar classes like benign keratosis (bkl) and melanocytic nevus (nv), these visualizations are crucial for deciphering the spatial focus of the model and comprehending failure cases.[Figure 9]

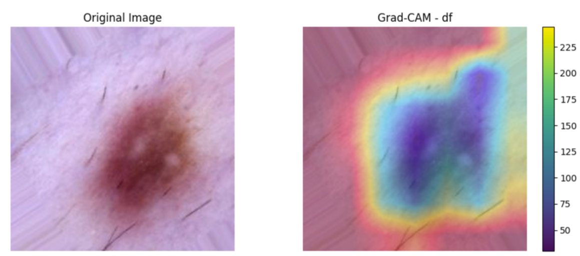
(a) Actinic keratosis (akiec)



(b) Basal cell carcinoma (bcc)



(c) Benign keratosis (bkl)



(d) Dermatofibroma (df)

Figure 8: Grad-CAM visualizations showing the model's focus on diagnostically relevant regions for (a) actinic keratosis and (b) basal cell carcinoma. The heatmaps highlight areas of clinical importance that contribute to the classification decision.

7 Conclusion

In Conclusion, to enhance performance and ability to be understood, this study proposed a deep learning method for classifying skin lesions using multi-scale feature fusion and soft attention mechanisms. The suggested model performed better than baseline CNN architectures in terms of accuracy and lesion localization through comprehensive testing, which included Grad-CAM and attention heatmaps. The model was able to focus on clinically important areas because of the attention modules, which improved the predictability and transparency of its results in a medical test.

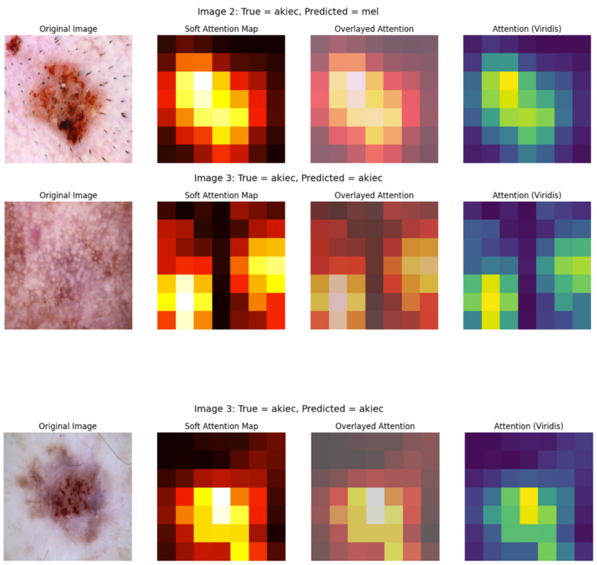


Figure 9: Attention heatmap visualization for a correctly classified and misclassified lesion.

8 Future Work

everal areas of improvement can be studied for further work. To improve evaluation accuracy, one approach is to combine image data with patient metadata such as age, sex, or lesion location, and the history of the family. The other option would be to implement the model as web-based online or mobile diagnosis applications, and evaluate its efficacy using different types of real-time clinical data. Moreover, analyzing selfsupervised learning strategies that use unlabeled data for supervised learning tasks would improve generalization even more, especially with imbalanced datasets. All things considered, this work establishes a baseline for creating high-performing, explainable artificial intelligence systems for diagnosis and other domains.

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